COURSE CODE: MLA03 **COURSE NAME:** Reinforcement learning for Environment and Agent

COURSE OUTCOME: CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

FACULTY: Dr. G. A. SENTHIL **ASSESSMENT WEIGHTAGE / MARKS:**

Assessment

Weightage: 30% | Total Marks: 50

Duration: 1 hr

Annexure A — Analytical Problem-Solving

Question 1: Warehouse Robot Optimal Path Planning using Dynamic Programming & Value Iteration (10 Marks)

Problem Statement: A warehouse robot must determine the shortest path to transport goods while avoiding static/dynamic obstacles. Apply Dynamic Programming to develop an optimal policy and explain how Value Iteration improves navigation efficiency.

Solution & Theoretical Formulation:

1. MDP Modeling for Warehouse Grid:

• **State Space (S):** Grid coordinates (x, y) representing robot locations, obstacles, and docking stations. • **Action Space (A):** Discrete movements {North, South, East, West, Stay}.

• **Transition Model P(s'|s, a):** Probability of reaching target cell s' given action a (deterministic in structured warehouses or stochastic with slip factor).

• **Reward Function R(s, a, s'):** +100 on reaching destination, -50 on obstacle collision, and -1 step cost per movement to encourage shortest path.

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| VALUE ITERATION ALGORITHMIC FLOW |
||
| [Initialize V(s) = 0 for all s in S, V(terminal) = 0] |
| | |
| v |
| [Loop: For each state s in S] |
| v_prev = V(s) |
| V(s) <- max_a sum_{s'} P(s'|s,a) [ R(s,a,s') + gamma * V(s') ] |
| delta <- max(delta, |v_prev - V(s)|) |
| | |
| v |
| [Check Convergence: Is delta < theta (Threshold)?] |
| | | |
| (No: Repeat Loop) (Yes: Converged) |
| | |
| v |
| [Extract Optimal Policy: pi*(s) = argmax_a sum_{s'} P(s'|s,a)[R(s,a,s') + gamma*V*(s')]] |
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```

2. Value Iteration Efficiency & Navigation Impact:

• **Bellman Optimality Backup:** Value iteration directly combines policy evaluation and policy improvement into a single step without waiting for exact policy evaluation convergence in each iteration. • **Global Path Optimality:** By updating the value function using the max operator over all legal actions, the robot avoids local minima and computes the shortest, obstacle-free trajectory to target drop-off zones.

Question 2: SARSA-Based Action-Value Learning for Autonomous Delivery Robots (10 Marks)

Problem Statement: A delivery robot operates in an uncertain environment with dynamic obstacles and slippery paths. Apply the SARSA algorithm to update the action-value function and explain how the learned policy adapts to environmental changes.

Solution & Theoretical Formulation:

1. SARSA On-Policy Learning Update Rule:

SARSA (State-Action-Reward-State-Action) is an on-policy TD control algorithm that updates the action-value function $Q(s, a)$ using the actual transition taken by the behavior policy:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)]$$

where α is the learning rate, γ is the discount factor, and A_{t+1} is selected using an ε-greedy policy: $A_{t+1} \sim \pi(\cdot|S_{t+1})$.

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| SARSA ON-POLICY INTERACTION CYCLE |
| |
| State S_t -----> Choose Action A_t using epsilon-greedy(Q) |
| | |
| v |
| Execute A_t in Environment -----> Receive Reward R_{t+1} and Next State S_{t+1} |
| | |
| v |
| Choose Next Action A_{t+1} from S_{t+1} using epsilon-greedy(Q) |
| | |
| v |
| Compute TD Target: Y_t = R_{t+1} + gamma * Q(S_{t+1}, A_{t+1}) |
| Compute TD Error: delta_t = Y_t - Q(S_t, A_t) |
| Update Q-Value: Q(S_t, A_t) <- Q(S_t, A_t) + alpha * delta_t |
| | |
| v |
| Set S_t <- S_{t+1}, A_t <- A_{t+1} and Repeat until Episode Termination |
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2. Environmental Adaptation & Safety Analysis:

• **On-Policy Safety Advantage:** Unlike Q-learning (which assumes greedy actions in the backup), SARSA accounts for the exploratory actions actually taken. In dangerous delivery routes (e.g., cliff paths or slippery terrain), SARSA learns safer paths that maintain clearance from hazards.

• **Dynamic Adaptation:** When physical terrain changes or obstacles move, negative rewards update $Q(s, a)$ locally, naturally shifting the ε-greedy policy to alternate viable detours.

Question 3: REINFORCE Algorithm for Continuous Robotic Arm Object Manipulation (10 Marks)

Problem

Statement: A multi-joint robotic arm performs continuous object grasping and manipulation. Apply the REINFORCE (Monte Carlo Policy Gradient) algorithm to develop a parameterized policy that maximizes task completion accuracy.

Solution & Theoretical Formulation:

1. Parameterized Continuous Policy & Likelihood Ratio:

Continuous joint torques $a \in R^d$ are generated by parameterizing the policy as a Gaussian distribution: $a \sim \pi_\theta(a|s) = N(\mu_\theta(s), \Sigma_\theta(s))$.

The policy gradient theorem defines objective optimization $J(\theta) = E_{\tau\sim\pi_\theta}[G_0]$ as:

$\nabla_\theta J(\theta) = E_{\tau\sim\pi_\theta} [\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) \cdot G_t]$

where return $G_t = \sum_{k=t}^T \gamma^{k-t} R_{k+1}$.

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| REINFORCE WITH BASELINE FOR CONTINUOUS ROBOTIC CONTROL |
| |
| 1. Collect Episode Rollout: tau = (s_0, a_0, r_1, s_1, a_1, ..., s_T) ~ pi_theta |
| | |
| v |
| 2. For each time step t = 0, 1, ..., T-1: |
| • Compute Discounted Return: G_t = sum_{k=t}^T gamma^{k-t} r_{k+1} |
| • Compute State Baseline Value: V_phi(s_t) |
| • Compute Advantage / Scaled Return: delta_t = G_t - V_phi(s_t) |
| | |
| v |
| 3. Accumulate Policy Gradient: |
| grad_theta J += grad_theta log pi_theta(a_t|s_t) * (G_t - V_phi(s_t)) |
| | |
| v |
| 4. Update Policy & Baseline Parameters: |
| theta <- theta + alpha_theta * grad_theta J |
| phi <- phi - alpha_phi * grad_phi || V_phi(s_t) - G_t ||^2 |
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2. Variance Reduction via Baseline:

Robotic manipulation has high degrees of freedom, leading to extreme variance in Monte Carlo returns. Incorporating a state-value baseline $V_\varphi(s_t)$ leaves the expected gradient unbiased while drastically reducing variance: $E[\nabla_\theta \log \pi_\theta(a_t|s_t) V(s_t)] = 0$.

Question 4: Comparative Analysis: Standard DQN, Double DQN (DDQN), and Dueling DQN (10 Marks)

Problem Statement: A complex game-playing AI and autonomous vehicle driving controller require value-based deep RL. Analyze the structural innovations and performance advantages of Double DQN (DDQN) and Dueling DQN over the standard DQN baseline.

Feature / Architecture Standard DQN (Mnih et al.) Double DQN (DDQN) Dueling DQN (Wang et al.)

Target Value Formulation $Y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ $Y = r + \gamma Q(s', \arg\max_{a'} Q(s', a'; \theta); \theta^-)$ $Y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ with dual streams

Core Architectural Modification Single deep neural network mapping state s to action values Q(s, a).

Overestimation Bias Mitigation Suffers from severe maximization bias (overestimates Q-values). **Convergence & Performance** Slow and prone to

oscillatory divergence in high dimensions. Decoupled action selection (online θ) from action evaluation (target θ^-).		identical long-term impact.
	Faster convergence with realistic and stable value bounds. Network bifurcates into State Value $V(s; \theta, \beta)$ and Advantage $A(s, a; \theta, \alpha)$.	Superior policy evaluation sample efficiency in complex state spaces.
Completely eliminates overestimation bias, improving policy stability.		
	Reduces variance in states where actions have	

Dueling Network Aggregation Identity: $Q(s, a; \theta, \alpha, \beta) = V(s; \theta, \beta) + [\, A(s, a; \theta, \alpha) - (1 / |A|) \sum_{a'} A(s, a'; \theta, \alpha) \,]$. Subtracting the mean advantage ensures unique identifiability of V(s) and A(s, a).

Question 5: Continuous Robotic Control (PPO/TRPO/DDPG) & Multi-Agent Traffic Coordination (10 Marks)

Problem Statement: Evaluate the suitability of PPO, TRPO, and DDPG for continuous robotic control, and analyze how Multi-Agent RL (MARL) coordinates smart city traffic management across scalability, non stationarity, and fairness.

Algorithm Key Mechanism / Constraint	**Sample Efficiency & Stability**	**Computational Complexity**
	Actor-Critic for continuous action deterministic policy $a = \mu_\theta(s)$ with replay buffer.	to hyperparameter tuning.
TRPO (Trust Region Policy Opt.) Enforces hard KL-divergence constraint: $E[D_{KL}(\pi_{old} \,\|\, \pi_\theta)] \leq \delta$ via Conjugate Gradient.	Highly stable; monotonic improvement guarantee. Moderate sample efficiency.	Very High (requires Hessian-vector products and Fisher Information Matrix).
		Low / First-order (uses standard SGD/Adam; easy to implement and scale).
PPO (Proximal Policy Opt.) Clipped surrogate objective: $L^{CLIP}(\theta) = E[\min(r_t(\theta)A_t, \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)A_t)]$.	High stability; superior empirical sample efficiency.	Moderate (susceptible to Q overestimation; addressed by TD3).
DDPG (Deep Deterministic PG) Off-policy	Sample efficient due to replay buffer, but sensitive	

Multi-Agent RL (MARL) for Smart Traffic Management Analysis:

• **Centralized Training with Decentralized Execution (CTDE):** Traffic light agents communicate during training via a centralized critic observing global traffic states, while executing decentralized policies at individual intersections using local sensor inputs.

• **Non-Stationarity Mitigation:** Because agents update policies simultaneously, the environment appears non stationary. CTDE and value-factorization methods (e.g., QMIX, MAPPO) stabilize joint convergence. • **Fairness & Ethical Objectives:** Reward functions balance total vehicle throughput with maximum wait-time thresholds to prevent traffic starvation on minor arterial roads.

Code of Conduct:

I **K. V. S. S. Jeswanth Babu/ 192472012** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

NAME: K. V. S. S. Jeswanth Babu REG NO: 192472012 **SIGNATURE OF THE STUDENT: K. V. S. S. Jeswanth Babu DATE:** 25-08-2026

RUBRICS FOR CALCULATION

Level 4

Level 3

Level 2

Level 1

Conceptual Understanding 25 Demonstrates

deep understanding of concepts with complete

clarity Explains concepts correctly with

minor gaps Partial understanding with errors or omissions

Unable to explain basic concepts

Application 25

Effectively applies concepts to new situations; solves problems

logically Applies concepts with minor errors Limited ability to apply

concepts; requires guidance Unable to apply concepts effectively

Analytical

Thinking 20 Critically analyzes scenarios with strong justification

and reasoning Provides reasonable analysis with some justification Superficial

analysis with weak reasoning No analytical reasoning demonstrated

Communication 15

Communicates ideas clearly, confidently, and with appropriate

terminology Communicates clearly but with minor hesitation Communication lacks

clarity or structure Unable to communicate ideas effectively

Response to Questions 15 Responds accurately and confidently, extending answers with depth Responds correctly

but with limited depth Responds partially; requires prompting Unable to respond to questions effectively